

# Sustainable Computing Hackathon 2026

(v1.0)

What you do is pretty much up to your team, as long as it's covering some element of sustainable computing. Below we give some rough ideas for groups to work on. We have grouped these into **themes**. At the start of the hackathon we will suggest people go to a theme of their choice to form a team to work on a project of their choice. Towards the end of the second day, teams can present their work and we aim to give out a small prize (more on this at the start of the second day!).

The introduction will highlight why we need to reduce the environmental impact of computing. The below ideas have a focus relating to the operational carbon footprint but we encourage groups to think of wider environmental concerns (water, minerals etc), and also to consider embodied emissions. (See <https://greencompute.uk/> and <https://rise.mmu.ac.uk/enrol/index.php?id=103> and <https://learn.greensoftware.foundation/>)

The outputs could be anything! You might want to build some form of application (e.g. a web app) or a mobile app handling a calculation, or an interactive website that helps users find relevant information (e.g. for their business). It's up to you, but if not sure then ask any of the mentors who will be circulating the rooms.

## Themes

We have grouped our ideas into three themes, using the below icons.

|                                                                                   |                                             |                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>Information and Education</b>            | apps (etc) that educate users (of a particular sector?) and/or provide information                                                                                                                                                                               |
|  | <b>Measuring/reducing energy</b>            | apps (etc) that either measure or estimate the energy consumed by a given activity, or approaches to reduce the energy consumption of a given activity                                                                                                           |
|  | <b>Measuring/reducing carbon footprints</b> | rather than considering energy, these projects would look at how to obtain the carbon footprint and/or how to reduce the carbon footprint. These projects would consider how green the grid is, or will be, rather than focusing on how much energy is consumed. |

Several projects suggested below could cover more than one theme, we've just chosen which we think is the main theme. And your project can do all of these themes or even none of them, as long as it's connected with sustainable computing!

**Please see the Resources document and our website for explanation of some terms and to see what's available to use / build upon**

## Find my carbon footprint

For a given type of application, calculate the energy consumption and lookup carbon intensity (CI) to determine carbon footprint. You could choose, say, something to do with email, or storage, or streaming,

Requirements: think which type of application; think how to calc the energy; and how to get the carbon footprint from the energy consumed (or directly from the supplier?); consider embedded as well as operational energy/carbon

**Alternatives:** as well as carbon could be other environmental footprints (water, minerals, impact on global south, ...); rather than focus on a single business application, this could be a survey of all the business' computing activities; consideration of Scopes 1,2,3

**Extensions:** rather than running here/now can you forecast where/when would give lower/lowest carbon footprint (given any business constraints)

## Quantify cost of cloud

Produce a calculator/app that gives best estimate of environmental costs of using Microsoft Azure GCP, AWS (or are there other cloud providers that give this direct

**Alternatives:** provide comparison of which cloud provider is best for a given type of use; quantify/critique cloud provider's estimate of Scope 3 emissions.  
(can use Real Time Cloud Project, Green Pixie for some data)

**Extensions:**

## Green Scheduling

For a given type of application, we would want it to run in a location and/or time such that the total carbon cost is significantly lower. Can you write a scheduler that would schedule a given business application to run in a different location and/or at a different time (in the future)

## AI

Carbon

Energy

Quantify the footprint of given AI tasks applicable to business (e.g. vibe coding, chatbot, agentic AI)

## Frugal AI

Info

How can we use AI without harming the planet?

## Students/University

Energy

How could you help MMU reduce the energy bill of its computing? Or what can you do at home/in your student residences?

## Promo pages (see company requirement)

Info

From our survey it seems some companies are unsure where to start and how to get hold of information relevant to them. Can you produce an app that works out what they need to know and then automates finding it?

## Quantify the environment cost of...

Info

People often ask what is the environmental cost of an email (or a full email box!) or a Teams chat, or how to compare saving a file versus using pen & paper, or whether to leave attachments in an email or to save to OneDrive. Your project could take such examples and work out the (comparative) environmental costs.